

Underwater Vision-Based Surveillance Using ML and AI

Day 6 — Methodology V1 & Updated Experiment Plan

1. Objective

To finalize the methodology, experiment plan, evaluation metrics and dataset strategy before model training.

2. Literature Review

The Day 2 literature review was completed. The shortlisted methods are:

- Classical enhancement: White Balance + CLAHE
- Learning-based enhancement: Water-Net, FUnIE-GAN
- Detection: YOLO-based transfer learning

3. Dataset Readiness

Three datasets will be used separately:

Dataset	Classes	Train	Valid	Test
DIATAquarium	11	8121	773	Not present locally
Aquatic Plant	1	1892	179	89
Well	4	3795	191	184

Note: Counts in this section are verified using the Day 7 inspection scripts.

DIATAquarium discrepancy: Day 4 estimated 9281 images, while Day 7 verified 8894 images in the available train/valid folders. The difference of 387 images will be re-confirmed with the mentor.

DIATAquarium YAML: The val and test paths both point to test/images, but no local test folder exists. This will be corrected before training so that val points to the existing valid/images folder.

4. Methodology V1

Dataset → Inspection → Preparation → Enhancement → YOLO Detection → Evaluation → Comparison

- Preserve the original dataset.
- Check image and annotation quality.

- Keep valid empty-label/background samples.
- Apply enhancement methods.
- Train/evaluate YOLO using the selected dataset splits.

5. Baseline

Original Image → YOLO → Detection Results

This provides the reference performance.

6. Experiment Plan

Experiment	Enhancement	Detector
01	None	YOLO
02	White Balance + CLAHE	YOLO
03	Water-Net	YOLO
04	FUnIE-GAN	YOLO

Experiments will be performed separately on all three datasets.

7. Evaluation

- Detection: Precision, Recall, mAP@0.5, mAP@0.5:0.95 and inference time.
- Enhancement: UIQM; PSNR/SSIM only when suitable reference images are available.

8. Split Strategy

- Train → model learning
- Validation → tuning/model selection
- Test → final evaluation when available
- Existing splits will be preserved where possible.
- Empty labels will be retained as possible negative/background samples unless later inspection proves otherwise.

9. Mentor-Confirmed Scope

- All 3 datasets will be used separately.
- Original vs Enhanced images will be compared.
- Detection metrics: Precision, Recall, mAP@0.5 and mAP@0.5:0.95.

- Enhancement metrics will be used where applicable.
- No video is currently available, so tracking is excluded from the current implementation.

10. Research Question

“How does underwater image enhancement affect the performance of deep-learning-based object detection across the three supplied underwater datasets?”